MNJ INSTITUTE OF ONCOLOGY &

REGIONAL CANCER CENTRE, RED HILLS, HYDERABAD.

DEPARTMENT OF NUCLEAR MEDICINE-PET CT

Patient Name: Mr.Lingam Rama Laxmi **Age / Sex: 47/F**

Date of Investigation: 26/06/2023 **Ref Doctor:** MNJIO&RCC

Hospital Reg No: 2023-00828 **PET Scan ID:** 1829/23

Clinical History: Known case of carcinoma right breast cT4bN1M0 on NACT with 6 cycles of TCH for residual evaluation.

^{18}F-FDG WHOLE BODY PET-NCCT STUDY

Procedure: *Sequential PET-CT with Whole body NCCT images followed by PET imagesacquired in 3-D mode ~45 minutes after injection of 10 mCi ^{18}F-FDG by intravenous route in hypoinsulinemic Euglycaemic state , from the level of orbits to mid-thigh including brain.*

FBS before injection was 115 mg/dl.

PET-CT FINDINGS: *The overall bio distribution of FDG is within normal physiological limits.*

HEAD, NECK & BRAIN:

-Note made of left frontal sinusitis and left sphenoid sinusitis.

-Mildly increased FDG concentration noted in enlarged right level Ib lymphnodes largest measuring 9mm with max SUV of 7.

-Non FDG avid subcentimetric bilateral level II nodes.

Physiologic FDG uptake is seen in the salivary glands, and larynx. Visualized paranasal sinuses, skull base do not show any abnormality on CT or any abnormal FDG uptake.

(18 F-FDG PET-CT has low sensitivity in detecting brain metastases due to low Signal to noise ratio as GLUT in brain are insulin independent0

THORAX & UPPER LIMBS:

-Intense FDG concentration noted in heterogenous soft tissue density lesion in the upper outer quadrant of right breast infiltrating overlying skin and focal loss of fat planes with underlying muscle measuring 4.4cm(AP)x4.4cm(ML)x4.2cm(CC) with max SUV of 21.

-Intense FDG concentration noted in heterogenous enlarged left axillary lymphnodes largest measuring 2cm(AP)x2.7cm(ML) with max SUV of 18.

There is no nodal hypermetabolism in the mediastinum. Bilateral lung parenchyma appears unremarkable. No evidence of pleural effusion or pleural thickening noted on either side. Physiologic FDG uptake is seen in the myocardium. Large airways, great vessels and other mediastinal structures appear normal on CT.

ABDOMEN-PELVIS:

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- **Mildly increased FDG concentration noted in enlarged right inguinal lymphnodes largest measuring 1.7cm with max SUV of 4.**

There is no nodal hypermetabolism in the abdomen or pelvis.The liver, gallbladder, pancreas, and spleen are normal. There are no adrenal nodules. Physiologic FDG excretion is seen in the kidneys and bladder.

MUSCULO-SKELETAL SYSTEM:

-Diffuse uptake noted in the marrow

Physiologic FDG distribution is seen in the visualized axial and appendicular skeleton.

IMPRESSION: Known case of carcinoma right breast cT4bN1M0 on NACT with 6 cycles of TCH for residual evaluation present scan reveals:

-RESIDUAL METABOLICALLY ACTIVE SOFT TISSUE DENSITY LESIONS IN UPPER OUTER QUADRANT OF RIGHT BREAST INFILTRATING OVERLYING SKIN AND UNDERLYING MUSCLE.

-RESIDUAL METABOLICALLY ACTIVE RIGHT AXILLARY LYMPHADENOPATHY

-NO OTHER METABOLICALLY ACTIVE LESIONS NOTED.

IN COMPARISON WITH PREVIOUS SCAN DATED 23/12/2022 FEATURES SUGGESTIVE OF PARTIAL METABOLIC RESPONSE AS PER PERCIST CRITERIA

Dr. RANADHEER MANTRI
PROFESSOR OF NUCLEAR MEDICINE